INCH-POUND

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6 March 1992
SUPERSEDING
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MILITARY SPECIFICATION

POSITIONERS, MELDING, TABLE TYPE, GEAR DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gear driven table type welding positioners of the floor mounted, powered table rotation design with the function of rotating workplaces for performing precision welding operations.

1.2 Classification. This specification covers welding positioners of the following classes and sizes. The class and size to be furnished shall be as specified (see 6.2).

Class A - Powered table rotation and powered table tilt.

Size 2	- 1,000	pounds	capacity.
Size 2a	- 1,500	pounds	capacity.
Size 3	- 2,000	pounds	capacity.
Size 4	- 2,500	pounds	capacity
Size 5	- 3,000	pounds	capacity.
Size 6	- 4,000	pounds	capacity.
Size 6a	- 5,000	pounds	capacity.
Size 7	- 6,000	pounds	capacity.
Size 8	- 10,000	pounds	capacity.
Size 8a	- 12,000	pounds	capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5213, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Size 9 - 16,000 pounds capacity.
Size 10 - 24,000 pounds capacity.
Size 11 - 30,000 pounds capacity.
Size 12 - 40,000 pounds capacity.
Size 13 - 50,000 pounds capacity.
Size 14 - 60,000 pounds capacity.
Size 15 - 70,000 pounds capacity.
Size 16 - 85,000 pounds capacity.

Class B - Powered table rotation and manual tilt.

Size la - 300 pounds capacity.
Size lb - 660 pounds capacity.
Size 2 - 1000 pounds capacity.
Size 3 - 1500 pounds capacity.
Size 4 - 2500 pounds capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specification, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters. (DOD adopted)

ANSI B5.1O - Machine Tapers. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363) .

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems. (DOD adopted)

NEMA M6 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete positioner shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The positioner shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The positioner shall be capable of supporting, rotating, and tilting workplaces for performing effective welding operations. The positioner shall include all components, parts, and features necessary to meet the performance requirements specified herein. Additional features which are not specified herein, but are a part of the manufacturer's current design, shall be included in the positioner furnished. Class A positioners shall be provided with powered table tilt. Class B positioners shall have manual tilt and sizes la, lb and 3 shall provide tilt action of from O to 90 degrees and sizes 2 and 4 shall be provided with tilt action of 135 degrees. All class A positioners shall provide table tilt action from O to 135 degrees of travel. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the positioner shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate rotation and tilt motion shall be graduated in increments not larger than five degrees. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.2.2 Reclaimed-materials. The positioner may contain reclaimed materials to the maximum extent possible, provided materials will not jeopardize the intended use, performance, or design life of the Positioner* Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

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3.2.3 Energy efficiency. The positioner and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, electrical and electronic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assembles furnished on the positioner, whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Covers, guards, or other safety devices. Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial positioner supplied to the commercial market, shall be provided for the point of operation and all other parts of the positioner that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration the procuring activity shall provide part design data to the supplier of the positioner (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the operation of the positioner. All guards shall allow access to the guarded part.

3.2.5.2 Noise level. Unless otherwise specified (see 6.2), noise emitted by the positioner when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response.

3.2.6 Environmental protection. During the manufacture, service, transportation, or storage of the positioner and its components, the positioner shall meet all applicable Environmental Protection Agency (EPA) and (OSHA) restrictions in effect on the date of the contract. The operation of the positioner at its final location shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.7 Mercury restriction. The positioner shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.8 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the positioner.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. Provisions shall be supplied for draining the reservoir for periodic cleaning.

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3.2.10 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or positioner.

3.3 Construction. The positioner shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the positioner to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the positioner shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the positioner and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have a hardness, surface finish, and toughness that will enable the gear train to transmit full-rated torque and horsepower of the drive motor without gear damage, failure, or premature deterioration and wear of the positioner.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The positioner shall consist of not less than a frame and base assembly, table, table drive system, motor controls, and components necessary to comprise a complete positioner system.

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3.4.1 Frame and base. The frame and base assembly shall be constructed of either cast steel or fabricated steel plate and shall be of sufficient strength and rigidity to maintain original alignment without distortion of the frame or workpiece while operating at maximum load condition. The frame and base assembly shall have provisions for securing the positioner to a mounting surface. The positioner shall be provided with a ground block located on the base or frame for the connection of an electrical ground cable.

3.4.2 Table. The workpiece table shall be constructed of either cast iron or fabricated steel plate and shall be of sufficient strength and rigidity to maintain alignment without distortion of the table or workpiece while operating at maximum capacity load conditions. Unless otherwise specified (see 6.2), the table work surface shall be machined flat to within +1/32 of an inch. Unless otherwise specified (see 6.2), the table shall be either round or square shape arid be not less than the dimensions specified herein. All table operations shall be smooth and shall not show evidence of galling or binding. Rotation shall be provided with stability and variability shall be as specified herein.

3.4.2.1 T-slots. Tables shall be provided with four equally spaced radial T-slot. The T-slots shall conform to ANSI B5.1 and shall be of the self-cleaning type. T-slot length on the square table shall be not less than 80 percent of the diagonal dimension of the table. Tables shall be provided with pilot holes and four matching T-nuts . When specified (see 6.2), round tables shall be supplied with concentric circles to assist in positioning parts on the table. The concentric circles shall be scribed every 2 inches of the radius. When specified (see 6.2), square tables shall be supplied with eight equally spaced T-slots plus eight T-slots adjacent and parallel to the table edges.

3.4.2.2 Spindle. When specified (see 6.2], a hollow spindle shall be furnished on all positioners. Spindle bearings shall be as required for continuous performance and shall be protected from ground welding current. The spindle taper and runout shall conform to ANSI B5.1O.

3.4.3 Table drive system. The table drive system shall be powered by a variable speed, reversible, silicon controlled rectifier (SCR) type, direct current (DC) drive system and a self-locking wormgear arrangement. Motors shall be provided with magnetic or dynamic brakes and shall conform to the requirements of 3.4.4.2. All positioners shall be provided with 360 degrees of power table rotation. All gears shall be adjusted for minimum backlash of .002 inch per inch of radius or less and, after adjustment, the gear case or adjustment point shall be secured to assure safety. Ground weld current conduction for all positioner systems shall be not less than 500 amperes for sizes 2 through 7 for class A positioners and not less than 1000 amperes for sizes 8 through 16. Ground weld current conduction for all positioner bearings movement systems shall be not less than 500 amperes for all sizes of class B positioners.

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TABLE I. Class A positioner, size and capacity.

Positioner size	Table load center of gravity and eccentricity, (distance from table top) (inches)	Table load capacity (pounds)	Table size (dia. or inches per side - square shape table) (inches)	Table variable speed range (RPM)	Table constant speed range (RPM)	Table tilt speed 0 to 135 degrees (seconds)	Table height range, flat, (inches) min. - max.	Table spindle through hole depth (inches)	Table centering hole depth (inches)	Table centering hole diameter (inches)
2	6	1,000	30	.05 to 2	2	15	18 to 50	1-3/8	1/2	1-1/2
2a	6	1,500	30	.08 to 1.3	1.3	24	32 to 50	1-3/8	1/2	1-1/2
3	6	2,000	30	.015 to 1.3	1.3	24	34 to 53	1-3/8	1/2	1-1/2
4	6	2,500	30	.07 to 1.5	1.5	24	34 to 53	2	1/2	2
5	6	3,000	30	.015 to 1.2	1.2	30	41 to 66	2	1/2	3
6	6	4,000	32	.031 to 1.2	1.2	46	41 to 66	2	1/2	3
6a	6	5,000	42	.031 to 1.2	1.2	46	42 to 66	2	1/2	3
7	12	8,000	46	.025 to 1.0	1.0	46	36 to 60	3	1/2	3
8	12	10,000	48	.020 to .80	.80	46	36 to 60	6	1/2	3
8a	12	12,000	60	.020 to .80	.80	46	49 to 73	6	1.0	9-1/8
9	12	16,000	60	.020 to .60	.80	60	49 to 73	7	1.0	9-1/8
10	12	24,000	60	.012 to .48	.48	87	59 to 83	7-1/2	1-1/4	10-1/4
11	12	30,000	66	.007 to .48	.48	87	65 to 85	7-1/2	1-1/4	11-1/8
12	12	40,000	72	.007 to .30	.30	100	65 to 85	8	1-1/4	11-1/8
13	12	50,000	84	.007 to .30	.30	100	65 to 85	12	1-1/4	11-1/8
14	12	60,000	84	.008 to .32	.32	100	65 to 85	14	1-1/4	11-1/8
15	12	70,000	96	.009 to .53	.53	104	72 to 81	15	1-1/4	11-1/8
16	12	85,000	96	.007 to .40	.40	104	72 to 81	16	1-1/4	11-1/8

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TABLE II. Class B positioner, size and capacity.

Positioner size	Table load-capacity (lbs)	Table size (dia. or side) sq. shape table (inches)	Table height	Table tilt capacity (deg)	Table variable speed range (RPM) min. - max.	Protractor increments (deg.)	Table load center of gravity (inches)	Table load eccentricity (inches)
1a	300	20	18	90	0.06 to 2.5	5	3	3
1b	660	17	21	90	0.23 to 2.3	5	6	1-1/2
2	1,000	30	36	135	0.2 to 2.0	15	6	6
3	1,500	30	36	90	0.33 to 1.3	15	6	6
4	2,500	30	36	135	0.15 to 1.5	15	6	6

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3.4.3.1 Table rotational system . Unless otherwise specified (see 6.2), the speed of table rotation for class A or B shall conform to tables I or II, be reversible, infinitely variable, and accurate within plus or minus 1 percent of the set speed adjustment. When specified (see 6.2), the rotational system shall be provided with a high speed device separate from the standard speed control, that will provide rapid rotation to maximum speed rate, without affecting the set rotational speed adjustment.

3.4.3.2 Power table tilting system. Class A positioners shall be provided with a power table tilting system. Powered table tilting shall be accomplished either by an SCR (DC) motor drive or a (AC) brake motor drive through a self-locking wormgear or gear reducer drive system. The system shall be provided with limit switches or sensors to prevent over-tilting at either extreme. The positioner shall be provided with a tilt protractor for accurate indexing of the tilt settings in increments of not greater than 5 degrees. Positioners, sizes 7 and larger, shall have dual gear segments with dual drive pinions. A braking system adjustment shall be provided with the system.

3.4.3.3 Manual table tilting system. Class B positioners shall be provided with a self-locking wormgear drive manual table tilting system. A handcrank shall be provided to facilitate infinitely variable, full range tilt operations. A protractor shall be provided to accurately indicate tilt settings in increments of 15 degrees or less. The tilt system shall be provided with positive tilt stops to prevent over-tilting in either extreme. The table for sizes la, lb and 3 positioners shall be capable of tilting from O to 90 degrees. Sizes 2 and 4 positioners shall be capable of tilting from O to 135 degrees.

3.4.4 Electrical requirements. The positioner shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each positioner shall be designed to operate on 3-phase, 60-Hertz (Hz), 230/460 dual voltage. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The positioner shall be initially wired as specified (see 6.2).

3.4.4.1 Motors. Motors shall meet the requirements of 3.4.4. The positioner motor shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.4.2 Magnetic reversing starter and safety switch. Unless otherwise specified (see 6.2), the positioners shall be furnished with a safety disconnecting switch and reversing magnetic starter, both with overload protection. Electronic type speed-drive controls shall be mounted in an enclosure conforming to NEMA ICS 1.

3.4.4.3 Controls. Unless otherwise specified (see 6.2), the class A and class B positioners shall be equipped with a pendant type control. When specified (see 6.2), a foot-operated control shall be furnished. The pendant control cable for class A positioners shall not be less than 15 feet long for

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sizes 2 through 7 and not less than 30 feet long for positioner sizes 8 through 16. The pendant control cable for class B positioners sizes 1a, lb and 3 shall be not less than 7 feet long, and not less than 15 feet long for positioners 2 and 4. The pendant control unit for each positioner shall have the following controls:

a. Table direction of rotation-forward, off, reverse selector switch.

b. Table rotation-start and stop buttons.

c. Table rotation-rapid traverse button.

d. Table rotational speed potentiometer.

e. Table tilt forward and reverse buttons (for class A positioners only).

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the positioner shall meet the requirements of table 1 or II for the class and size positioner specified. Positioner class, size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2.), the physical size of the positioner shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Table speed accuracy requirements. Table speed for class A or B positioner shall be accurate within the requirements specified in 3.4.3.1.

3.7 Performance. The positioner shall be capable of rotating and tilting capacity workpiece loads as specified in table I or II for the type and size machine specified. Sizes 2 through 6a of class A positioners shall be capable of rotating and tilting capacity loads specified in table I with the center of gravity of the workpiece located 6 inches above the table and 6 inches eccentricity from the rotation center. Sizes 7 through 16 of class A positioners shall be capable of rotating and tilting capacity loads specified in table I with the center of gravity of the workpiece located 12 inches above the table and 12 inches eccentricity from the rotation center. Class B positioners shall be capable of rotating and tilting capacity loads specified in table II with the center of gravity of the workpiece and eccentricity from the rotation center as specified.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial positioner shall be furnished.

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3.9 Optional and accessory equipment. Optional and accessory equipment shall be furnished as specified and fully described (see 6.2). The equipment listed below shall be functional without requiring modification of the equipment or positioner for the class and size specified.

Three jaw self centering chuck.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on charts, and plates. All words on the machine including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant

3.11.1 Lubrication plate or chart. When specified (see 6.2), a lubrication plate or chart shall be attached to each positioner. If a chart is furnished, it shall be place: in a transparent plastic folder or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2), the

Points of lubricant application
Type of lubricant
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each positioner. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Table speed range (RPM)
Table tilting speed in seconds (IPM)
Positioner load capacity (pounds)
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.12 Workmanship Workmanship of the positioner and its accessories shall meet all requirements specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The positioner shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each positioner prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.10. The positioner shall pass the examination, all tests, and the inspection to be accepted.

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4.6 Acceptance test. An acceptance test shall be performed on each positioner to ensure specific conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the positioner is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each positioner shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 degrees to 104 degrees Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.8.1.1 Operational test. The positioner shall be placed on a suitable surface and connected to the appropriate power supply. The positioner shall be operated for not less than 30 minutes. The positioner shall be operated through all table speed ranges and shall be examined to determine compliance. Table rotation shall be maintained during tilting testing. All pendant or panel mounting controls and indicators shall be operated and examined to determine compliance. Table tilt time to operate from flat position to 135 degrees shall not exceed the time allowance cited in table I for class A positioner. Tilt limit switches shall be inspected and tested for compliance. Manual tilt positioners shall be operated over the full range of travel and the mechanical stops shall be inspected for compliance. Proper operations of all positioner functions, controls, motors, adjusting mechanisms, and accessories shall be verified for proper operation during the trial period.

4.8.2 Table speed accuracy test. Positioners shall be tested to determine compliance with the accuracy requirements of 3.4.3.1. The table speed accuracy requirement test shall be performed prior to and subsequent to the performance test given in 4.8.3. Any positioner component adjustments required to perform and comply with 3.4.3.1 and 3.6 during the performance test shall be cause for rejection.

4.8.3 Performance test. A capacity load in accordance with table I or II based on the class and size positioner specified shall be secured to the table surface. With the table-positioned ‘over the frame and base, the positioner shall be adjusted to the maximum vertical height without the use of counterweights, and shall be operated through all rotational angle movements. All positioners shall be secured to a mounting surface before the operational and performance tests are performed. All table operations shall be smooth and shall not show evidence of galling or binding. Table rotation shall be inspected for speed stability and range shall be in compliance with table I or II. Positioners with vertical adjustment shall be tested for operational movement. DC amperage equal to or not less than the ground weld current ampers specified for the class and size positioner shall be passed through the positioner with the table rotating for a period of or not less than one minute with no current interruption or damage to the positioner. All performance tests shall be completed without making any adjustments to or performing any maintenance on the positioner.

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4.8.3.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8.3 shall be as specified and fully described (see 6.2).

4.8.3.2 Power and manual table tilt test. The class A positioner shall be connected to a power source with a capacity load. The class A positioner shall be power cycled through all tilting speeds and gear sequential increments specified in table I and 3.4.3.2. Cycling from 0 through 135 degrees, the power tilting of the table shall be accomplished through all tilting increments in seconds as specified in table I not less than three times, and tested, inspected, and verified that the table tilting system meets or exceeds the requirements specified herein. The class B positioner shall be connected to a power source with a capacity load. The class B positioner shall be manually cycled through all degrees of tilting and gear sequential increments specified in 3.4.3.3. The positioner shall be cycled from O through 90 degrees for sizes la, lb and 3, and 0 to 135 degrees for sizes 2 and 4. Manual tilting of the table shall be accomplished through all tilting increments in degrees as specified in table II not less than three times. The positioner shall be tested, inspected, and verified that the table tilting system meets or exceeds the tilting requirements specified in 3.4.3.3 herein. All performance tests shall be complete without making any adjustments to or performing any maintenance on the positioner.

4.8.3.3 Noise level test. Noise level measurements shall be made to ensure conformance with 3.2.5.2. Unless otherwise specified (see 6.2), the measurements shall be made at the operator's work station and at each "corner of the positioner. Noise emitted by the positioner when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type II sound meter at slow response. The test shall be conducted with the positioner operating at intermediate and maximum speeds.

4.8.3.4 Electromagnetic interference control test. Equipment requiring electromagnetic Interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the positioner offered complies with all environmental protection restrictions as required in 3.2.6. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.10 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirement for packaging shall be in accordance with MIL-W-45562.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Class A welding positioners are used to position and rotate workplaces for welding in downhand operator positions and where extensive tilting of the weldment is required. Class B positioners are used to position and rotate workplaces manually tilting of the workpiece.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound) (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5.1).

h. Noise level requirements, if different (see 3.2.5.2).

i. Environmental protection requirements exceeding EPA/OSHA requirements, if different (see 3.2.6).

j. Specify surface finish requirements, if different (see 3.3.3).

k. Painting, if different (see 3.3.5.).

1. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

m. Table shape, if different (see 3.4.2).

n. Machined table face, if different (see 3.4.2).

o. Scribed table, if required (see 3.4.2.1).

p. Eight equally spaced T-slots plus eight slots adjacent and parallel to sides, if required (see 3.4.2.1).

q. Table spindle, if different (see 3.4.2.2).

r. Table rotation speed, if different (see 3.4.3.1).

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s. Separate high speed drive system, if required (see 3.4.3.1).

t. Voltage requirements, if different (see 3.4.4).

u. Initial voltage, as specified (see 3.4.4).

v. Motors, if different (see 3.4.4.1).

w. Magnetic reversing starter, if different (see 3.4.4.2).

x. Controls, if different, (see 3.4.4.3).

y. Foot operated control, if required (see 3.4.4.3)

z. Size and capacity, if different (see 3.5).

aa. Specify physical size limitation, if required (see 3.5.1).

bb. Additional performance features, if required (see 3.7.1).

cc. Specify and fully describe optional equipment, if required (see 3.9).

dd. Specify applicable repair parts, if required (see 3.10).

ee. Lubrication plate, if required (see 3.11.1).

ff. Lubrication plate information, if different (see 3.11.1).

gg. Nameplate, if different (see 3.11.2).

hh. Inspection conditions, if different (see 4.3).

ii. First article inspection, if different (see 4.4).

jj. Quality conformance inspection, if different (4.5).

kk. Acceptance test, if different (see 4.6).

ll. Additional performance tests, if required (see 4.8.3.1).

mm. Noise level acceptance test, if different (see 4.8.3.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-805Z7	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 501O.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

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MIL-P-80068C	MIL-P-80068D
Class A	Class A
Size 1	Size 1 (deleted) Size 2a (new size) Size 8a (new size) Size 15 (new size) Size 16 (new size)
Class B	Class B
Size 1	Size 1 (deleted) Size 1a (new size) Size 1b (new size) Size 3 (new size) Size 4 (new size)

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

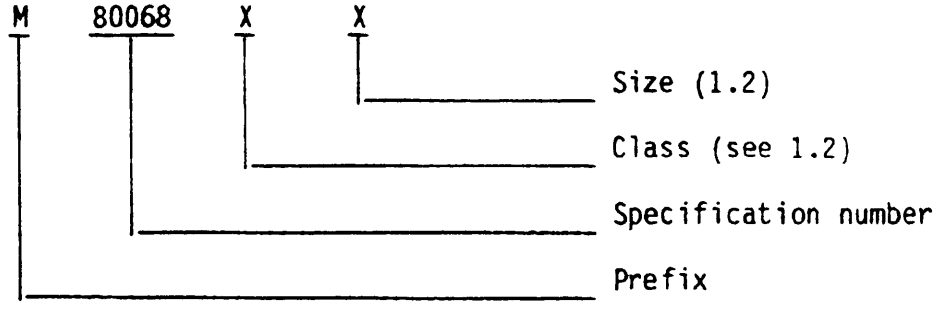
6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PIN to be used for positioners acquired to this specification is created as follows:



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6.14 Subject term key word listing.

Downhand welding
Rotating workpiece support
Positioner, welding
Tilting workpiece support
Welding parts holder
Welding support

Custodians:
Air Force - 99
Navy - SH

Review activities:
Army - AR
Navy - AS
Air Force- 84
DLA - GS

User activity:
Navy - MC

Preparing Activity:
DLA - IP

(Project 3436-0049)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-80068D	92 02 06

3. DOCUMENT TITLE

POSITIONERS, WELDING, TABLE TYPE, GEAR DRIVEN

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	(1) Commercial (901)-775-6699	(2) AUTOVON 683-6699
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5213	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete. 198/270

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